The world would be a dull place if all particles were free. Forces are what make particles do interesting things, such as assembling themselves into atoms, molecules, chocolate bars, and black holes. The force on any given particle is the sum total of the forces exerted on it by all the other particles in the universe. In practice, we usually assume that we know what all the other particles are doing and replace their effect by a potential energy function for the particle that we are studying. This much is true in both classical and quantum mechanics.

The potential energy function is denoted by $V(x)$. In classical mechanics it's related to the force on a particle by the equation

$$F(x) = -\frac{\partial V}{\partial x}.$$

If the motion is one-dimensional, the partial derivative can be replaced by an ordinary derivative, but I will leave it as is. If we then combine this equation with Newton's second law, $F = ma$, we get

$$m\frac{d^2x}{dt^2} = -\frac{\partial V}{\partial x}.$$

In quantum mechanics, we proceed differently; we write a Hamiltonian and solve the Schrödinger equation. Incorporating the potential energy into this program is straightforward. The potential energy $V(x)$ becomes an operator $\mathbf{V}$ that gets added to the Hamiltonian.

What kind of operator is $\mathbf{V}$? The answer is easiest to express if we think in the language of wave functions rather than in terms of abstract bras and kets. When the operator $\mathbf{V}$ acts on any wave function $\psi(x)$, it multiplies the wave function by the function $V(x)$.

$$\mathbf{V}|\Psi\rangle \;\rightarrow\; V(x)\psi(x).$$